

Estimating and Testing the Proportion of J. S. Bach Chorales Whose Soprano Ends on Scale Degree 1 at the Final Cadence

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1 Data and Model

The data come from the 371 Riemenschneider Bach chorales, four-part harmonizations of Lutheran hymn tunes composed by Johann Sebastian Bach in the first half of the eighteenth century and collected in the 1941 Riemenschneider edition. The data were accessed as Craig Sapp’s Humdrum `kern` encodings (Sapp, n.d.) and parsed in base R. Following De Clercq (2015), who identified 24 duplicate harmonizations and one non-Bach chorale (no. 150, harmonized by Johann Rosenmüller), these were removed, keeping the lower-numbered member of each duplicate pair. Three additional chorales (nos. 3, 82, 262) were excluded because two independent key-assignment methods disagreed. The working sample consists of $n = 371 - 1 - 24 - 3 = 343$ unique chorale harmonizations.

The unit of observation is one final cadence per chorale. The success event is $X_i = 1$ if the soprano pitch at the final cadence is scale degree 1 in the tonic key, measured by letter distance from the tonic, and 0 otherwise. The tonic key was determined from the key signature together with the final bass note, with the key designation encoded in the source files (including modal designations such as Dorian) used as a fallback. Chorales with Phrygian or other modal melodies, whose final cadences conventionally land on scale degree 5, are included as non-successes. We model $X = \sum_{i=1}^n X_i \sim \text{Binomial}(n, p)$, where p is the true proportion of chorales whose final soprano note is scale degree 1.

Notation:

- p : true population proportion of final soprano scale-degree-1 endings
- $\hat{p}$: sample proportion estimator
- $X = 316$: number of successes in the sample
- $n = 343$: sample size
- $p_0 = 0.90$: null hypothesis value
- $\alpha = 0.05$: significance level for the test and interval

Assumptions:

1. *Independence:* By taking one final cadence per chorale and removing duplicate harmonizations, we treat each chorale as an independent observation. Pooling several phrase endings from the same chorale would count the same compositional decision repeatedly.
2. *Identical distribution:* Each chorale has the same probability p of ending with soprano on scale degree 1.
3. *Fixed sample size:* $n = 343$ is treated as fixed by the corpus rather than random.

These assumptions are reasonable: after deduplication the chorales are distinct compositions, and there is no clear reason to expect dependence between different chorales. The identical distribution assumption is a simplification, since modal-melody settings structurally end off the tonic, but it is not unbelievable at the scale of the corpus.

2 The Estimator and Its Properties

The population parameter of interest is p , the true proportion of Bach chorales whose soprano ends on scale degree 1 at the final cadence. The estimator is the sample proportion:

$$\hat{p} = \frac{X}{n} = \frac{316}{343} \approx 0.9213.$$

Bias:

$$E[\hat{p}] = E\left[\frac{X}{n}\right] = \frac{np}{n} = p.$$

Therefore, $\text{Bias}(\hat{p}) = E[\hat{p}] - p = 0$. The estimator is unbiased.

Variance:

$$\text{Var}(\hat{p}) = \text{Var}\left(\frac{X}{n}\right) = \frac{\text{Var}(X)}{n^2} = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}.$$

Mean Squared Error:

$$\text{MSE}(\hat{p}) = \text{Var}(\hat{p}) + [\text{Bias}(\hat{p})]^2 = \frac{p(1-p)}{n} + 0^2 = \frac{p(1-p)}{n}.$$

Consistency: As $n \rightarrow \infty$, $\text{Var}(\hat{p}) = p(1-p)/n \rightarrow 0$. By Chebyshev's inequality (Casella and Berger, 2002), for any $\epsilon > 0$,

$$P(|\hat{p} - p| > \epsilon) \leq \frac{\text{Var}(\hat{p})}{\epsilon^2} = \frac{p(1-p)}{n\epsilon^2} \rightarrow 0.$$

Thus $\hat{p} \xrightarrow{p} p$, so the estimator is consistent.

Point estimate and interpretation: The estimated proportion is $\hat{p} = 0.9213$, with plug-in standard error $\sqrt{\hat{p}(1-\hat{p})/n} = 0.0145$, meaning approximately 92.1% of the 343 unique Bach chorales end with the soprano on scale degree 1 at the final cadence. This suggests that ending on the tonic soprano note is overwhelmingly the norm in Bach's chorale harmonizations, at a rate slightly above the 90% benchmark tested below.

3 Interval Estimation

We construct a 95% confidence interval for p , so $\alpha = 0.05$.

Clopper-Pearson (exact) interval:

$$(\text{Beta}(\alpha/2; X, n - X + 1), \text{Beta}(1 - \alpha/2; X + 1, n - X)),$$

where $\text{Beta}(q; a, b)$ is the q -quantile of the Beta distribution. This interval is derived by inverting the binomial test and has guaranteed coverage $\geq 1 - \alpha$ in finite samples, i.e., exactly correct (conservative) coverage rather than only asymptotically correct coverage. By contrast, asymptotic intervals such as the Wald interval can have coverage below the nominal level when p is close to 1 (Brown, Cai, and DasGupta, 2001).

Computed limits:

$$(0.888, 0.948).$$

Interpretation: We are 95% confident that the true proportion of Bach chorales ending with the soprano on scale degree 1 lies between 0.888 and 0.948. The interval contains $p_0 = 0.90$, so the data are compatible with the convention-based benchmark even though the point estimate exceeds it.

4 Hypothesis Testing

Null and Alternative Hypotheses:

$$H_0 : p = 0.90 \quad \text{versus} \quad H_1 : p > 0.90.$$

A one-sided alternative is used because the question of interest is whether Bach's practice exceeds the 90% benchmark; there is no substantive interest in whether the proportion is less than 90%. The null value $p_0 = 0.90$ is sourced from De Clercq (2015), who found 166 of 177 major-key chorales (93.8%) end with a perfect authentic cadence with the soprano on the tonic; 0.90 is a conservative rounding used as the convention-based null.

Test Statistic and Null Distribution: Under H_0 , the test statistic is $X \sim \text{Binomial}(n = 343, p_0 = 0.90)$. The exact p-value is:

$$p\text{-value} = P(X \geq 316 \mid p_0 = 0.90) = \sum_{x=316}^{343} \binom{343}{x} (0.90)^x (0.10)^{343-x} = 0.108.$$

Type I Error Control: The exact binomial test has type I error $\leq \alpha$ in finite samples due to the discreteness of the binomial distribution. The normal-approximation z-test has only asymptotic type I error control; for comparison, it gives $z = 1.314$ with one-sided p-value 0.094, in close agreement with the exact test here.

Interpretation: At $\alpha = 0.05$, since the p-value $0.108 > 0.05$, we fail to reject H_0 . The observed rate of 92.1% is above the benchmark, but the evidence is not sufficient to conclude that the true proportion exceeds 90%.

5 Discussion

The analysis shows that approximately 92.1% of Bach’s chorale harmonizations end with the soprano on scale degree 1, with a 95% exact confidence interval of (0.888, 0.948) that contains the 90% convention-based value. This finding is close to De Clercq’s (2015) major-key figure of 93.8% but extends it by providing formal statistical inference, using a programmatic definition, and taking one final cadence per chorale as the unit of observation. The two definitions do not fully agree: De Clercq’s major-context settings of the “Herzlich tut mich verlangen” melody end with tonic-key cadences whose soprano is on scale degree 3, which count as tonic endings under his cadence-based definition but not under the melodic definition used here. As a check, applying the same definition to all 2,112 fermata phrase endings gives a phrase-level proportion of 0.361; the corresponding interval would be artificially narrow because phrase endings within the same chorale are correlated, which is why the chorale is used as the unit of observation here. Limitations include the three excluded chorales with conflicting key evidence and the reliance on encoded keys where the signature-plus-bass method fails. Future work could model all phrase endings using clustered methods (e.g., mixed-effects or GEE) to use more data while accounting for within-chorale dependence. The analysis code and full run output are available on request.

References

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